



CHAPTER I

INTRODUCTION

1 INTRODUCTION

The automotive service industry relies heavily on efficient after-sales operations, as the quality of service delivery directly affects customer satisfaction and overall service experience (Terre & Almario, 2024; Gupta, 2022). Customer handling becomes increasingly challenging when service processes lack consistent monitoring and control, particularly as service volume grows and operational activities become more complex (Velimirovic, 2022). In many service environments, incomplete documentation, delayed job status updates, and decentralized records lead to poor coordination and slower service completion, highlighting the need for integrated garage or service management systems (Patidar, 2021; Beken et al., 2024).

Inventory control constitutes a critical factor, as stock monitoring and parts availability directly influence turnaround time and operational costs (Barcelona et al., 2024; Baylosis et al., 2023). Forecasting spare parts demand enhances planning by reducing uncertainty and guiding stocking decisions based on historical data (Fan, 2023). Additionally, decision support systems convert operational data into reports and insights that enable managers to monitor performance, identify trends, and make informed decisions (Kozin, 2023; Teerasoponpong & Gupta, 2022). In response to these challenges, this capstone project aims to design and develop a centralized web-based Automotive Service Operations and Decision Support System for a multibranch automotive/highway tire service center, integrating service management, inventory tracking, forecasting, and reporting to improve operational efficiency and strengthen management oversight (Roque et al., 2021; Chukwumuanya, 2024).

1.1 Background of the Study

Automotive and Highway Tire Service Centers conduct continuous daily transactions requiring coordination among service advisors, technicians, and inventory personnel. Typical activities include customer reception, job order creation, vehicle inspection, documentation of findings, technician assignment, work recording, parts tracking, service status updates, and preparation of operational reports for management. In multi-branch setups, these activities become more difficult to control when each branch maintains separate records or relies on manual documentation, resulting in scattered information that is challenging to consolidate.

Inconsistent or decentralized service documentation causes employees to spend additional time locating previous service histories, validating prior repairs, confirming replacement parts, and tracking job progress. This results in repeated data entry, delayed updates, missed service details, and decreased efficiency in customer handling, especially when customers seek follow-up service at different branches. Management also faces challenges in consolidating service performance, monitoring turnaround time, identifying bottlenecks, and comparing branch outputs when reports are manually generated or based on incomplete data. Inventory management further affects daily operations because service completion depends on the availability of parts, tires, lubricants, and other consumables. If inventory tracking is not



linked to service transactions, stock movement becomes harder to monitor, resulting in inaccurate counts, stock inconsistencies, delayed replenishment, and unnecessary overstocking. Since service and parts demand varies, forecasting becomes important for predicting future needs, improving replenishment planning, and minimizing service delays caused by unavailable parts.

To address these operational gaps, this study proposes a centralized web-based system integrating service transaction recording, customer and vehicle profiling, inventory monitoring, forecasting, and decision support reporting. Centralizing data and standardizing key workflows across branches aims to enhance service coordination, improve monitoring, and provide management with improved visibility and data-driven decision support.

1.2 Objectives of the Study

General Objective

To design and develop an Automotive Service Operations and Decision Support System for multi-branch service centers that integrates service management, inventory tracking, forecasting, and reporting to enhance operational efficiency and managerial decision-making.

Specific Objectives

- To design and develop a centralized web-based system supporting customer and vehicle profiling with searchable service history across branches.
- To develop a service management module for job order creation, service transaction recording, inspection findings documentation, and tracking of service workflow stages (queued, ongoing, completed, released).
- To implement documentation of work performed, detailing specific repair, maintenance, and corrective actions executed by technicians based on findings and approved job orders.
- To develop a parts and materials usage feature directly linked to inventory deduction to enhance traceability of item consumption.
- To implement an inventory stock monitoring module that tracks availability, usage, and movement per branch, incorporating alerts for critical stock levels.
- To integrate a forecasting module that analyzes historical service and inventory data to estimate future demand and support stock planning decisions.
- To provide decision support dashboards and automated reports for branch-level and consolidated monitoring of service volume, inventory usage, repeat visits, and turnaround time. To implement role-based and branch-based access control with activity logging to ensure secure and accountable system usage.



- To test the developed system to verify that each module performs its intended function correctly.
- To develop a user's guide to support proper and consistent system usage.

1.3 Significance of the Study

This study is significant because it aims to develop a centralized and data-driven Automotive Service Operations and Decision Support System that strengthens how a multi-branch automotive/highway tire service center records, monitors, and manages service transactions and inventory in one platform. Integrating service documentation, inventory tracking, forecasting, and decision-support reporting is expected to improve accuracy of records, accessibility of information, timeliness of monitoring, and the quality of managerial decisions.

Service Center Owners and Managers.

The system provides a consolidated view of operations across branches through dashboards and operational reports. This supports improved monitoring of service volume, turnaround time, repeat visits, and inventory usage. Additionally, it enables management to identify operational issues earlier, compare branch performance, and implement improvements based on evidence-based insights.

Service Advisors.

The system enhances customer handling by organizing job orders, inspections, service updates, and service history within a single searchable platform. This reduces time spent retrieving records, improves documentation accuracy, and standardizes service processing across branches.

Technicians.

The system improves coordination by providing clear access to assigned work orders, inspection findings, required parts, and service status updates. Documented work performed and progress tracking reduce miscommunication and ensure tasks are completed and properly recorded.

Inventory Custodians and Purchasing/Planning Staff.

The inventory module enhances visibility of stock availability and movement per branch, helping to reduce shortages and overstocking. The forecasting function further supports planning by guiding replenishment decisions based on historical service and parts consumption patterns.

Customers.

The system can contribute to faster and more consistent service by minimizing documentation delays and supporting better parts availability planning. Access to complete service history across branches can also strengthen continuity of service and follow-up transactions.

Future Researchers and System Developers.



This study serves as a reference for future projects involving multi-branch service operations, integrated inventory and forecasting features, and decision support reporting. The approach and results may guide similar studies and future enhancements.

1.4 Scope and Limitation

Scope of the Study

This study encompasses the design, development, testing, and evaluation of a web-based centralized management system for a multi-branch automotive/highway tire service center. The system centralizes service operations by enabling branches to record service transactions and maintain customer and vehicle service histories in a shared database. It supports multibranch coordination through branch-based visibility and managerial access for consolidated monitoring.

The system scope includes:

Service Management: job orders, inspection findings, documentation of work performed, service status tracking, and service history recording;

Customer and Vehicle Records (CRM-oriented): customer profiles, vehicle profiles, and cross-branch service history retrieval;

Inventory Monitoring: stock availability, usage, movement per branch, and linking service transactions to stock deduction;

Forecasting: statistical forecasting using historical service and inventory data to estimate future demand and support stock planning;

Decision Support and Reporting: dashboards and automated reports for monitoring service volume, inventory usage trends, repeat visits, and turnaround time across branches;

Access Control and Security: role-based and branch-based permissions with activity logging for accountability.

The system is implemented using PHP for the backend, MySQL for the database, and HTML/CSS for the frontend. It is accessed through a standard web browser and designed for use in branch offices.

Limitations of the Study

The study is limited to the operational processes of a multi-branch automotive/highway tire service center; thus, organizations with different workflows may require adjustments in system design and implementation. As the system is web-based, its performance and accessibility



depend on stable internet or local network connectivity, and disruptions may affect crossbranch access and real-time updates.

Forecasting results produced by the system are estimates dependent on the completeness and accuracy of historical service and inventory data stored in the database. Missing, inconsistent, or limited historical data may reduce the reliability of forecasting outputs. The system focuses on service operations, inventory monitoring, forecasting, CRM-oriented records, and decision support reporting. It excludes full accounting modules (e.g., payroll, tax computation, full bookkeeping) and advanced integrations such as supplier automation, automated purchasing, diagnostic hardware integration, and SMS/email marketing automation.

1.5 Definition of Terms

The following key terms are operationally defined according to their specific use in this study. These definitions are intended to provide clarity and establish a common understanding of the concepts central to the research, particularly those involving service management, customer and vehicle records, inventory tracking, forecasting, and decision support within a multi-branch automotive/highway tire service center.

Activity Logging. A feature that records user actions performed in the system, such as creating, editing, or updating records, to promote accountability and security.

Automotive Service Operations. The daily activities involved in managing automotive or highway tire services, including customer reception, job order processing, vehicle inspection, repair documentation, service status monitoring, and release of completed work.

Automotive Service Operations and Decision Support System. The proposed web-based system in this study that integrates service management, inventory monitoring, forecasting, and reporting to improve operational efficiency and support managerial decision-making in a multi-branch service center.

Branch-Based Access Control. A system feature that restricts users' access to records and functions based on their assigned branch, ensuring that branch personnel can only manage data relevant to their location.

Centralized System. A system in which data from all branches are stored and managed in one shared database, allowing easier access, monitoring, and consolidation of records.

Customer Profiling. The process of recording and maintaining customer information, such as personal details and transaction history, to support better service handling and follow-up transactions.

Dashboard. A visual display of important operational information, summaries, and performance indicators used by managers to monitor service activities, inventory status, and branch performance.



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Decision Support System (DSS). A computer-based system that collects, processes, and presents data in the form of reports, summaries, or insights to help managers make informed decisions.

Forecasting. The process of using historical service and inventory data to estimate future demand for parts, materials, or services in order to support planning and replenishment decisions.

Inspection Findings. The recorded observations, issues, or conditions identified during the checking and assessment of a vehicle before service work is performed.

Inventory Monitoring. The process of tracking stock availability, usage, and movement of parts, tires, lubricants, and other materials used in service operations.

Job Order. A formal service record containing the details of the customer request, vehicle condition, findings, assigned work, parts needed, and service status.

Managerial Decision-Making. The process by which managers analyze operational information and reports to plan actions, solve problems, and improve branch performance.

Multi-Branch Service Center. An automotive or highway tire service business operating in more than one branch or location under one management system.

Parts and Materials Usage. The recording of items consumed during service operations, which are directly linked to stock deduction in the inventory system.

Repeat Visits. Instances in which customers return for additional or follow-up service, which may be monitored to evaluate customer retention and service quality.

Role-Based Access Control. A security feature that limits system access according to the responsibilities of a user, such as manager, service advisor, technician, or inventory custodian.

Service History. A collection of previous service records of a customer or vehicle, including job orders, findings, repairs performed, and parts used across branches.

Service Management Module. The part of the system responsible for handling job orders, service transaction recording, findings documentation, work performed, and service status updates.

Service Status Tracking. The monitoring of the current stage of a service transaction, such as queued, ongoing, completed, or released.

Shared Database. A common database used by all branches to store and retrieve unified records for customers, vehicles, services, and inventory.



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Stock Movement. The flow of inventory items in and out of storage, including stock additions, deductions, and transfers.

Turnaround Time. The amount of time required to complete a service transaction, from job order creation up to the release of the vehicle.

Vehicle Profiling. The process of recording and maintaining vehicle information, such as plate number, model, and service records, to support accurate tracking and service continuity.

Web-Based System. A system that is accessed through a web browser over a network, allowing authorized users to use its functions online within the service center environment.

Work Performed. The specific repair, maintenance, or corrective actions carried out by technicians based on inspection findings and approved job orders.